

A Ditlevsen–Karppinen transition-density competitor for the Dacca cholera benchmark

Computational note

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1 Question and evaluation target

Reviewer 1 proposed constructing tractable Gaussian transitions using the strong-order-1.5 approximation of Ditlevsen and Samson (DS), then using those transitions in SAEM, Fisher-identity gradient estimation, PGAS, or a particle Gibbs method stabilized by Karppinen, Singh, and Vihola (KSV). The statistical criterion in the DMOP paper is not the complete-path pseudo-likelihood. It is the log likelihood of the Euler-discretized POMP, independently evaluated by a bootstrap particle filter. Accordingly, a candidate method is successful only if its estimated parameter vector has a competitive Euler-model log likelihood.

The paper used 100 searches from the same wide box, compared methods under similar elapsed time, evaluated selected parameter vectors with 5,000 particles and 36 replications, and emphasized both the right tail and median of the final log likelihoods. Its best comparable-effort result is -3744.17 for IFAD-0.97; extended IFAD-0.97 reaches -3744.19 with median -3744.22 .

2 Dacca state representation

For the benchmark, $c = \alpha = 1$, $\rho = Y_0 = 0$, and $\delta = 0.02$ are fixed. Thus the inapparent-infection coordinate remains zero. In population-fraction coordinates, the active state is

$$X = (S, I, R^1, R^2, R^3, M)^\top \in \mathbb{R}^6,$$

where M accumulates disease deaths within a month and is reset after each measurement. With drift $b_\theta(X, t)$, the single diffusion vector field is

$$g_\theta(X, t) = \sigma SI(-1, 1, 0, 0, 0, 0)^\top, \quad dX_t = b_\theta(X_t, t)dt + g_\theta(X_t, t)dB_t.$$

The observation model is exactly the manuscript model, $Y_n \mid X_{t_n} \sim \mathcal{N}(P_{t_n} M_{t_n}, \tau^2 P_{t_n}^2 M_{t_n}^2)$.

3 Diffusion rank, transition rank, and bracket depth

It is essential not to confuse the two ranks. Hypoellipticity means that the *instantaneous diffusion matrix is not full rank* while the finite-time transition nevertheless has a smooth density. DS assume $V \in \mathbb{R}$, $U \in \mathbb{R}^p$, and

$$C(v, u) = \begin{bmatrix} 0_{1 \times p} \\ \Gamma(v, u) \end{bmatrix}, \quad \Gamma(v, u) = \text{diag}\{\sigma_1(v, u), \dots, \sigma_p(v, u)\}, \quad \sigma_j > 0.$$

Thus C has rank $p < p + 1$ by construction. The directly forced rough block already supplies p directions and the first drift bracket supplies the one missing smooth direction. The order-1.5 term is exactly what gives their leading approximate covariance a nonzero variance of order Δ^3 in that smooth coordinate for sufficiently small Δ .

Equivalently, the DS instantaneous diffusion has rank $d - 1$ in a d -state system. Dacca has $d = 6$ active coordinates but diffusion rank one, hence codimension five rather than one. Diffusion rank is invariant under an invertible change of state coordinates, so state dependence of the coefficient σSI does not put Dacca into the DS model class.

Dacca is potentially hypoelliptic too, but it is not in this depth-one form. Its active state has dimension six and its instantaneous diffusion has rank one away from $SI = 0$ and rank zero on that boundary. At the manuscript parameter and state point, the normalized columns formed with the Stratonovich drift $b^S = b - \frac{1}{2}(Dg)g$,

$$g, [b^S, g], [b^S, [b^S, g]], \dots$$

gain numerical ranks 1, 2, 3, 4, 5, 6 only through bracket depth five. This is evidence along one repeated drift-bracket chain that the true finite-time SDE transition can be full-dimensional; it is neither a proof of global hypoellipticity nor an argument that the diffusion should be full rank.

The local order-1.5 approximation still does not contain all six directions. For one Brownian driver, let L_0 be the Itô generator and let $L_1 f = (Df)g$. Write

$$A = L_1 b, \quad C = L_0 g, \quad D = L_1 g, \quad E = L_1 L_1 g.$$

The random part of the scalar-noise strong Taylor step can be written

$$Z_h = g\eta + A\xi + C(h\eta - \xi) + \frac{1}{2}D(\eta^2 - h) + \frac{1}{6}E(\eta^3 - 3h\eta), \quad (1)$$

$$\text{var}(\eta) = h, \quad \text{var}(\xi) = h^3/3, \quad \text{cov}(\eta, \xi) = h^2/2. \quad (2)$$

All terms in (1) have mean zero. With $u = g + \frac{h}{2}(A + C)$ and $v = A - C$, their exact moment covariance is

$$Q_h^{(1.5)} = hu u^\top + \frac{h^3}{12}vv^\top + \frac{h^2}{2}DD^\top + \frac{h^3}{6}EE^\top. \quad (3)$$

For additive noise, $C = D = E = 0$, and (3) reduces to the covariance used by DS; for their harmonic oscillator it is exactly their equation (34). The implementation retains all four terms because Dacca has multiplicative noise. It uses the matching second-order mean

$$m_{\theta, h}(x) = x + hb_\theta(x) + \frac{h^2}{2}\mathcal{L}_\theta b_\theta(x).$$

For Dacca, $g(x) = q(x)e$ with $e = (-1, 1, 0, 0, 0, 0)^\top$. The vectors C, D, E are therefore collinear with g , while $A = (Db)g$ supplies at most one further direction. Hence $\text{rank}\{Q_h^{(1.5)}\} \leq 2 < 6$. This concerns the approximate finite-step covariance, not the rank-one instantaneous diffusion. DS prove nondegeneracy for a $1 + p$ system with p directly forced directions; that argument does not by itself cover Dacca's six-state, one-driver system.

4 Which time scale is being compared

One Euler step per measurement interval is indeed the measurement time scale. The important distinction is between transition *schemes* on that same grid. With r substeps per month,

$r = 1$: one monthly DS transition, $r = 20$: twenty DS transitions of length 1/20 month.

DS demonstrate inference with their discretization points at the observation times. Introducing $r - 1$ intermediate latent states per month is an extension. In every case below, goodness is assessed against the *same* numerical model used by the DMOP manuscript: the Euler POMP with 20 substeps per month. A one-step Euler likelihood is therefore neither the target nor evidence about the one-step DS fit.

Grid equality also says nothing about numerical stability. The published Dacca point has a three-stage immunity rate $3\epsilon = 57.3$ per year. Applied to a decay mode with rate λ , the DS second-order conditional mean has stability polynomial $R(z) = 1 - z + z^2/2$, $z = h\lambda$, whose real-axis absolute-stability interval is $0 \leq z \leq 2$. The monthly, half-monthly, and five-substep values of $h(3\epsilon)$ are respectively 4.78, 2.39, and 0.96. Thus failure at $r = 1, 2$ and viability beginning at $r = 5$ are the expected coarse-step behavior of this mean, not evidence that the measurement and one-step grids were different.

5 Higher-bracket completion as a diagnostic

To test the strongest nearby density-based idea, define the local controllability expansion

$$A_q(x, h) = \left[h^{1/2}g, \frac{h^{3/2}}{1!}J_b g, \dots, \frac{h^{q-1/2}}{(q-1)!}J_b^{q-1}g \right], \quad (4)$$

$$(H_q)_{ij} = \frac{1}{i+j+1}, \quad Q_h^{(q)}(x) = A_q H_q A_q^\top. \quad (5)$$

This is the covariance obtained by integrating the Taylor expansion of $\exp(J_b s)g$ over a step. For $q = 2$ it reproduces the DS leading covariance; for $q = 6$ the unscaled Dacca local controllability matrix is generically full rank. The mean remains the DS second-order generator expansion. This “DS-Krylov” completion is a new local-linear Gaussian approximation, not a claim about the scope of the DS paper. At fine h , its smallest directions scale as high odd powers of h , so the transition becomes extremely ill-conditioned even when it is algebraically full rank. It is retained as a rank diagnostic and is not the transition used in the main experiment.

6 Monthly block transition

The executable method applies the order-1.5 approximation at each of the 20 internal steps and integrates out the 19 intermediate states before running SMC. For a month beginning at x , set $\mu_0 = x$ and $P_0 = 0$. Holding the forcing fixed at its value for internal step j , compute

$$\mu_{j+1} = m_{\theta, h}(\mu_j), \quad (6)$$

$$F_j = \left. \frac{\partial m_{\theta, h}(z)}{\partial z} \right|_{z=\mu_j}, \quad (7)$$

$$P_{j+1} = F_j P_j F_j^\top + Q_{\theta, h}^{(1.5)}(\mu_j), \quad j = 0, \dots, 19. \quad (8)$$

Equation (8) is a time-varying linear-Gaussian moment closure around the propagated mean. Conditional on that linearization, the intermediate states integrate out exactly, giving the monthly approximation

$$\tilde{p}_\theta^{\text{block}}(x_{n+1} | x_n) = \mathcal{N}\{x_{n+1}; \mu_{20}(x_n), P_{20}(x_n)\}.$$

Although every local $Q^{(1.5)}$ has rank at most two, the transported terms $F_{19} \cdots F_{j+1} Q_j F_{j+1}^\top \cdots F_{19}^\top$ need not share a two-dimensional range. Their sum can therefore give a full-dimensional monthly

covariance. In finite precision it is extremely ill-conditioned. The code adds 10^{-12} times its largest marginal variance to the diagonal once, after the 20 steps; no ridge is added within a step.

The bootstrap filter samples and scores one endpoint per month, applies the death-count measurement density there, and resamples. Its latent path thus contains monthly endpoints only. KSV bridge backward sampling is not needed for these runs because the 19 internal states are not part of the SMC path. Increasing the number of internal steps changes the transition approximation and its cost, but it does not lengthen the particle genealogy. This block variant therefore does not test the fine-grid path-degeneracy mechanism that motivates KSV.

The earlier extended-Kalman quasi-likelihood optimizer and the local-nugget particle fit are not used as headline DS results. At every parameter iteration, the executable block method runs the monthly bootstrap filter, draws one endpoint path from its forward genealogy, and takes a projected stochastic Fisher-score step. Because the Dacca complete pseudo-likelihood has not been shown to have DS’s curved exponential-family form, this is stochastic gradient ascent rather than the M-step in DS Algorithm 2. Every candidate is judged afterward by the original Euler-20 particle filter.

More precisely, write $\tilde{p}_\theta^{\text{block}}$ for the monthly Gaussian pseudo-model with latent endpoint path X . Fisher’s identity gives

$$\nabla_\theta \tilde{\ell}^{\text{block}}(\theta; y) = \mathbb{E}_\theta \left[\nabla_\theta \log \tilde{p}_\theta^{\text{block}}(X, y) \mid y \right].$$

At iteration m , forward SMC approximates $\tilde{p}_{\theta_{m-1}}^{\text{block}}(X \mid y)$ and one complete genealogy $X^{(m)}$ is drawn from its terminal weighted empirical measure. Automatic differentiation is then applied to $\log \tilde{p}_\theta^{\text{block}}(X^{(m)}, y)$ while holding $X^{(m)}$ fixed. Thus

$$\hat{z}_m = \nabla_\theta \log \tilde{p}_\theta^{\text{block}}(X^{(m)}, y) \Big|_{\theta=\theta_{m-1}}$$

is a one-path SMC approximation of the block pseudo-model observed score. It is not an estimate of the original Dacca SDE/Euler score, whose transition density is unavailable. No derivative is taken through particle simulation, resampling, or ancestor selection. Updating $s_m = (1 - a_m)s_{m-1} + a_m \hat{z}_m$ and stepping along s_m is a Robbins–Monro score method. In contrast, proper SAEM updates a parameter-independent sufficient statistic $S(X^{(m)}, y)$ and maximizes its surrogate exactly (or with a justified generalized-EM step).

7 Why KSV is relevant, and why it is not a direct patch

At $r = 1$ there are no artificial within-month latent points, so KSV does not solve the monthly Taylor-approximation or rank-depth issues. It can become relevant when DS is extended to $r > 1$ and the intermediate states are retained in the particle path. In ordinary particle backward sampling, the ancestor weight contains $p_h(x_{t+h} \mid x_t)$. As h decreases, this density concentrates and almost all backward weight is assigned to the existing ancestor. Increasing the number of Euler steps between monthly observations therefore degrades path mixing. KSV replace unit-step backward sampling by bridge updates spanning a fixed physical-time block. Their method directly targets this failure mode.

There is an important algorithmic distinction. DS Algorithm 2 draws a fresh path from an ordinary forward SMC approximation at every SAEM iteration; it does not require a conditional-particle-filter Markov kernel. KSV specifically stabilize the backward update of a CPF/particle-Gibbs kernel. Consequently KSV is not required for the monthly block competitor: its internal states have already been integrated out. It would become relevant if those states were explicitly imputed with particle Gibbs or another CPF path update.

KSV Assumption 7 requires evaluation of multi-step densities $M_{u|\ell}(x_u | x_\ell)$ and simulation from bridge conditionals $M_k(x_k | x_{k-1}, x_u)$. The nonlinear, state-dependent DS local transition does not supply these objects. A valid combination must introduce a tractable time-varying linear-Gaussian proposal M and put the density correction into the Feynman–Kac potentials. KSV also does not repair the rank-two DS transition, and Dacca’s complete likelihood is not the curved exponential family required for DS’s closed-form SAEM M-step. One must either derive a valid generalized-EM surrogate or use a separately labeled score method.

Input: observations $y_{1:N}$; initial θ_0 ; Euler step h ; particles J ; SA gains a_m ; physical-time block endpoints $\mathcal{T}$.

1. Construct the higher-bracket completed transition

$$p_{\theta,h}^K(x_k | x_{k-1}) = \mathcal{N}\left(m_{\theta,h}(x_{k-1}), Q_{\theta,h}^{(6)}(x_{k-1})\right).$$

2. At the current reference path, construct a globally time-varying linear-Gaussian proposal $M_{\theta,1:T}$ whose multi-step marginals and bridge conditionals are available by Kalman recursions.
3. Define correction potentials

$$G_{\theta,k}(x_{k-1}, x_k) = \frac{p_{\theta,h}^K(x_k | x_{k-1})}{M_{\theta,k}(x_k | x_{k-1})} \times g_\theta(y_n | x_k) \mathbf{1}_{\{k=t_n\}}.$$

4. For $m = 1, 2, \dots$:

- (a) Run a conditional particle filter for $(M_{\theta_{m-1}}, G_{\theta_{m-1}})$.
- (b) Apply KSV bridge backward sampling over $\mathcal{T}$, proceeding from the final block to the first and using the Gaussian bridge conditionals of M , to obtain $X_{0:T}^{(m)}$.
- (c) Compute the complete-data score $z_m = \nabla_\theta \{\log p_\theta(X_0^{(m)}) + \sum_k \log p_{\theta,h}^K(X_k^{(m)} | X_{k-1}^{(m)}) + \sum_n \log g_\theta(y_n | X_{t_n}^{(m)})\}$.
- (d) Update the stochastic approximation $s_m = (1 - a_m)s_{m-1} + a_m z_m$.
- (e) Take a constrained stochastic-score parameter step, e.g. $\theta_m = \Pi_\Theta \{\theta_{m-1} + \eta_m \text{Adam}(s_m)\}$.

Output: $\hat{\theta}$ and an independent Euler-particle-filter estimate of $\ell_{\text{Euler}}(\hat{\theta})$.

Algorithm 1: Full DS–Krylov–KSV stochastic-approximation score ascent (design, not the executed baseline)

Algorithm 1 is the minimal coherent combination of the two papers for Dacca. It is materially more than “put KSV inside DS”: it requires a higher-order rank completion, an auxiliary Gaussian bridge model, exact importance correction for that proposal, a forward conditional-particle pool, and a numerical M-step.

The code implements a narrower fixed-lower/fixed-upper blocked-interior conditional SMC kernel derived from the bridge construction in KSV Algorithm 8. Its auxiliary bridge conditionals and p^K/M corrections are internally consistent, but it cannot reconnect the upper endpoint to a different lower-boundary forward particle. It is therefore neither full Algorithm 8 nor CPF–BBS, and its fraction of changed interiors is not KSV’s probability of lower-boundary update (PLU). Short Dacca checks found mixed effects on parameter progress and no repair of the one-ancestor forward

genealogy. It is excluded from the headline baseline unless a full forward CPF–BBS implementation passes a separate invariance and mixing validation.

The executed numerical score method is a Fisher-identity Robbins–Monro extension, rather than the closed-form curved-exponential-family SAEM used by DS. Neither it nor the blocked-interior diagnostic is used to claim that full Algorithm 1 has been run.

References

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